



On-Demand Services Platform for Alamein

Solution Report | Sprint 3 Final Submission | ACIE Smart City Hackathon
Alamein International University | Smart Buildings & Built Environment Theme

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1. Executive Summary

Aoun is a hyper-local, mobile-first marketplace that digitizes the informal service economy of Alamein by connecting residents with verified skilled workers in real time. Built for the ACIE Smart City Hackathon under the Smart Buildings and Built Environment theme, Aoun tackles a specific, validated gap: the complete absence of a trusted digital channel between service seekers and service providers in Alamein's rapidly growing residential population.

The platform enables customers to search, compare, and book workers with full price transparency and a community rating system — while giving workers a formal digital identity to grow their client base and income. Aoun's AI-powered matching engine (Gemini API) surfaces the best available worker for each request, reducing search time from hours to seconds.

2. Problem Statement

The city of Alamein lacks a structured digital marketplace connecting residents with skilled local service providers. Residents are forced to rely on inefficient word-of-mouth methods, leading to opaque pricing, substandard outcomes, and wasted time. Meanwhile, qualified workers have no formal channel to advertise their services, resulting in 3 to 5 idle days per week and significant lost income.

This dual failure — information asymmetry on the demand side and invisibility on the supply side — is a classic market failure that digital platforms have solved in major cities worldwide, but no solution has yet been built for smaller, emerging Egyptian cities like Alamein.

Validated Pain Points

Pain Point	Affected Group	Evidence / Source
No way to find or compare local services	Residents	Team survey: 78% rely on asking neighbors
Overpriced services due to zero competition	Residents	Estimated 40-60% markup over fair price
No digital presence for skilled workers	Workers	0 formal platforms currently serve Alamein

3-5 idle days per week due to low visibility	Workers	Interviews with local workers, March 2025
Safety risk from unverified strangers	Residents	No rating/verification system exists today

3. Solution — How Aoun Works

Aoun is a three-sided platform connecting customers, workers, and the community through three core interactions: search, compare, and connect. The platform is mobile-first, Arabic-language first, and built specifically for the Alamein community context — making it meaningfully different from any global alternative.

Core User Journey

Step	Customer Action	Worker Action	Platform Role
1. Search	Selects service category	Profile visible to all nearby users	AI ranks best matches
2. Compare	Views workers, ratings & prices	Rating reflects past job quality	Transparent pricing layer
3. Book	Confirms booking in one tap	Receives push notification	Escrow & confirmation
4. Complete	Rates worker & pays in-app	Earns income, builds reputation	Releases payment + review

Key Differentiators

- **Hyper-local focus:** Built exclusively for Alamein — not a general platform stretched to fit. Community trust is built into the product DNA.
- **AI-powered matching:** Gemini API integration recommends the optimal worker based on service type, location, rating, and availability.
- **Arabic-first UX:** Designed in Arabic, with a UI/UX flow tailored to local expectations and interaction patterns.
- **Worker formalization:** Workers gain a verified digital identity, helping them build reputation over time and access more work.
- **Zero upfront cost for workers:** Commission-only model means workers take zero risk to join the platform.

4. Competitive Landscape

Several on-demand service platforms exist globally and regionally. None of them address the Alamein context, small-city trust dynamics, or Arabic-first UX at a hyper-local level:

Platform	Region	Model	Gap vs. Aoun
TaskRabbit	USA / Europe	Freelance marketplace	No Egypt presence; no Arabic UX
Khidma	Cairo, Riyadh	Home services app	Metro-only; no small-city model
Fixit Egypt	Cairo / Giza	Repair services	Urban-only; no community trust layer

Handyman.com.eg	Cairo	Directory listing	Static listings; no booking or ratings
Aoun	Alamein (first)	Hyper-local marketplace	Hyper-local, AI-matched, Arabic-first

5. Impact Projections

Impact estimates are grounded in publicly available data on comparable platforms, Alamein's known residential population, and conservative adoption curves consistent with campus-first rollouts at other Egyptian universities.

Metric	Year 1 Target	Year 3 Target	Basis
Active users	5,000+	25,000+	Alamein resident & student population
Worker income increase	~30%	~45%	Comparable platforms: 25-40% uplift
Time saved per booking	2-4 hours	2-4 hours	vs. word-of-mouth (consistent)
Jobs formalized	200+ workers	1,000+ workers	Informal sector digitization rate
Platform GMV (gross)	EGP 500K+	EGP 5M+	Avg booking EGP 300 x volume
SDGs addressed	SDG 8, SDG 11	SDG 8, 10, 11	Decent work + sustainable cities

6. Business Model

Aoun operates a commission-based marketplace model. The platform takes no upfront fees — revenue is earned only when a transaction completes, aligning platform incentives with worker and customer success. A premium listing tier (Phase 2) will allow workers to boost visibility during peak demand periods.

Block	Details
Value Proposition	Transparent pricing · Verified workers · Instant booking · Community ratings · AI matching
Customer Segments	Alamein residents & AIU students (demand) + Local skilled workers (supply)
Revenue Streams	10-15% commission per completed booking · Premium worker placement (Phase 2)
Key Activities	Platform development · Worker onboarding & ID verification · Community trust-building · AI tuning
Key Resources	Mobile app · Gemini AI integration · Worker database · Community reputation system
Channels	Mobile app (iOS & Android) · AIU campus network · WhatsApp referral · Word-of-mouth
Cost Structure	Cloud infrastructure · Development team · Worker onboarding ops · Marketing & growth
Key Partners	Alamein International University · Local worker communities · Gemini API (Google)

7. Go-to-Market Strategy

Aoun's GTM follows a phased rollout designed to build trust before scaling. By starting with the AIU campus community — a captive, tech-savvy, and geographically concentrated population — Aoun can iterate quickly, collect real usage data, and build a credible track record before expanding to the wider city.

Phase	Timeline	Target Audience	Key Actions & Milestones
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Phase 1 Campus Pilot	Month 1-3	AIU students + 50 local workers	Manual worker onboarding · Free tier · Feedback loops · 100 bookings
Phase 2 City Rollout	Month 4-9	All Alamein residents	Commission model live · Marketing push · Worker verification · 1,000 us
Phase 3 Expansion	Month 10-18	New Mansoura, New Cairo	Franchise / white-label model · Partner with other universities · 10K use

8. Technical Architecture

Aoun is engineered as a cloud-native, mobile-first application. The team has adopted a modern, production-ready stack that prioritizes fast iteration during the hackathon while remaining scalable for real-world deployment. The architecture separates concerns cleanly across frontend, backend, AI, and DevOps layers.

Layer	Technology / Tool	Responsibility	Owner
Mobile Frontend	React Native + V0 UI	Cross-platform iOS & Android app	Hamdy Sameh
Backend API	Node.js / Express	REST API, business logic, auth	Ahmed Saleh
AI Matching Engine	Gemini API (Google)	Smart worker recommendations	Mohamed Abdelhady
Cloud & DevOps	Containerized deployment	CI/CD pipeline, cloud infra, scaling	Ahmed Fekry
Product & Strategy	Platform architecture	Vision, roadmap, stakeholder management	Adham Ashraf
UI/UX Prototyping	V0 by Vercel	Design system and component library	Hamdy Sameh

9. Risks & Mitigation

Every marketplace faces supply-demand bootstrapping challenges and trust barriers. Aoun has identified the key risks and built specific mitigations into the launch plan:

Risk	Likelihood	Impact	Mitigation Strategy
Cold start: not enough workers at launch	High	High	Manual onboarding of 50 workers pre-launch via AIU network
Low customer trust in unverified workers	Medium	High	ID verification + community ratings from Day 1
Low smartphone adoption among older workers	Medium	Medium	Simplified onboarding; phone-based booking fallback
Competition from national platforms entering Alamein	Low	Medium	Community moat: hyper-local trust is hard to replicate
Payment / escrow regulatory constraints	Low	High	Launch with cash-on-completion; integrate fintech in Phase 2

10. Conclusion

Aoun addresses a genuine, unmet need in one of Egypt's newest and fastest-growing cities. The informal service economy in Alamein is large, active, and completely undigitized — making it an ideal first market for a platform that combines hyper-local trust, AI-powered matching, and a frictionless mobile experience.

The team brings together product strategy, full-stack engineering, DevOps, and AI capabilities under a clear vision: to become the digital backbone of Alamein's local economy — and then export that model to every new Egyptian city that faces the same challenge.

With SDG alignment on Decent Work (SDG 8) and Sustainable Cities (SDG 11), a viable commission-based business model, and a working prototype ready for campus pilot, Aoun is positioned to deliver measurable social and economic impact from day one.

Elevator Pitch (Day 2): "Alamein's residents can't find or trust local service workers — and those workers have no digital way to reach customers. Aoun fixes this with a hyper-local marketplace that connects both sides instantly, transparently, and for the first time."